W-S-871b

June 6, 1963

SUPERSEDING
Interim Fed. Spec. W-S-00871a (GSA-FSS)
December 11, 1962 and
Fed. Spec. W-S-871
May 7, 1935

FEDERAL SPECIFICATION

SWITCH; KNIFE, OPEN, FRONT- AND REAR-CONNECTED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers single- and multi-pole, open, knife switches, with or without fuseholders and bases, and designed for either front or rear wiring connection.

1.2 Classification.

1.2.1 *Types, classes, and styles.* Knife switches furnished under this specification shall be of the following types, classes, and styles, as specified (see 6.1).

Type I.—Heavy-duty switch of milled and machine-made built-up construction.
Type II. — Medium-duty switch of punched and bent-up parts.

Class 1,—Fusible.
Class 2.—Nonfusible.

Style A.—Front connected, with individual base.
Style B.—Rear connected, without base (type I only).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

PPP-B-566—Boxes, Folding Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New

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York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods of.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Underwriters' Laboratories, Inc., Publication:

UL-363—Standard for Knife Switches.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or 1655 Scott Blvd., Santa Clara, California.)

3. REQUIREMENTS

3.1 Except as otherwise specified herein, switches furnished under this specification shall conform to the applicable requirements of Underwriters' Laboratories, Inc., UL-363 Standard for Knife Switches.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the knife switches he proposes to supply under this specification conform to the requirements of Underwriters' Laboratories, Inc., UL-363 Standard for Knife Switches. The label, or listing with reexamination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the knife switches conform to this requirement.

3.2.2 In lieu of the label, or listing, with reexamination, the bidder may submit independent proof satisfactory to the contracting agency that his switches conform to the applicable requirements of the published standards, including methods of tests, of the Underwriters' Laboratories, Inc., UL-363, Standard for Knife Switches.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Ratings.

3.3.1 *Voltage.* The standard voltage ratings for knife switches shall be 125, 250, and 600 volts a.c./d.c., 250 volts d.c.—500 volts a.c., and 500 volts a.c.

3.3.2 *Current.* The standard current ratings for knife switches shall be 30, 60, 100, 200, 400, 600, 800, 1200, 1600, 2000, 2500, 3000, 4000, 5000, and 6000 amperes.

3.4 General.

3.4.1 All copper used for current-carrying parts except terminal connector lugs and rear-connected stud nuts shall be of not less than 98 percent conductivity. (International annealed copper standard.)

3.4.2 Single lugs shall be used for all capacities from 60 to 600 amperes, inclusive. Single or multiple lugs may be used for capacities of 800 and 1,200 amperes. Rear-connected switches of 1,500 amperes capacity and over shall be furnished without lugs.

3.4.3 Cast wire terminal lugs shall have the same inside dimension as wrought-copper lugs, except that the metal shall be of such additional thickness as will carry continuously, the rated current without a rise

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in temperature in excess of 30 degrees C. (86 degrees F.)

3.4.4 The minimum depth of terminal-lug cable hole shall be 1-1/2 times the diameter.

3.4.5 Front-connected fused knife switches with switch and fuse parts in tandem shall be of the high-jaw construction.

3.4.6 Front-connected switches, designed for double throw and having three or more poles, shall not have front-connected terminals for the hinge contacts of the inner poles, unless standard switch spacings between adjacent live metal parts of opposite polarity are secured either by increased spacing between poles or by the use of barriers.

3.4.7 Rear-connected knife switches up to and including 1,200 amperes rated capacity shall be furnished with one lug per terminal stud.

3.5 Terminal studs, rear-connected switches.

3.5.1 Maximum length of laminated studs on rear connected switches shall be as follows:

Amperes	Maximum length in inches
1,600	8
2,000	8-1/2
3,000	10
4,000	10
6,000	10
8,000	10
10,000	10

3.5.2 Sizes of slots in laminated studs for rear-connected switches shall be 1/4 inch.

3.5.3 The minimum area of contact surface for laminated studs in rear-connected switches shall be based on the following densities.

Amperes	Amperes
1,600 to 3,000 incl.	150 per square inch.
3,001 to 6,000 incl.	125 per square inch.
6,001 to 10,000 incl.	100 per square inch.

3.5.4 On all round studs of rear-connected switches, one nut and one washer for clamping the contact to the base, and two nuts for contacting the terminal lug or bus-bar shall be furnished.

3.6 Performance. Switches furnished under this specification shall comply with the requirements for heating, overload, and dielectric strength of UL-363, Standard for Knife Switches.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Unless otherwise specified (see 6.1), sampling, inspection, and tests of switches furnished under this specification shall conform to the provisions of UL-363 Standard for Knife Switches.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.)

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified.

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5.1.1 *Level A.* Each switch shall be individually preserved and packaged in accordance with MIL-P-116, method IC-2.

5.1.2 *Level B.* Each switch shall be individually packaged in a box conforming to PPP-B-566, style II, type A or B, class a; PPP-B-665, style optional; or to PPP-B-676, type optional. The boxes shall be closed in accordance with the applicable specification or appendix thereto.

5.1.3 *Level C.* The switches shall be packaged in accordance with the suppliers' commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified.

5.2.1 *Level A.* The switches shall be packed in boxes conforming to PPP-B-636, class 2, V3c, style optional. Each box shall be waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* The switches shall be packed in boxes conforming to PPP-B-636 type I, class 1. Each box shall be sealed and closed in accordance with the appendix to the box specification.

5.2.3 *Level C.* The switches shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with rules and regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Type, class, and style of switch (1.2.1).
(*c*) Voltage and current rating of switch (3.3).
(*d*) Number of poles required.
(*e*) Administrative provisions for inspection records (4.1).
(*f*) Additional sampling, inspection, or test procedures, if required (4.2).
(*g*) Special marking, if required (5.3).
(*h*) Level of packaging and packing required (5.1 and 5.2).

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

6.3 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Switches, electric, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:
Switches, electric, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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MILITARY INTEREST:

Army—DESC

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